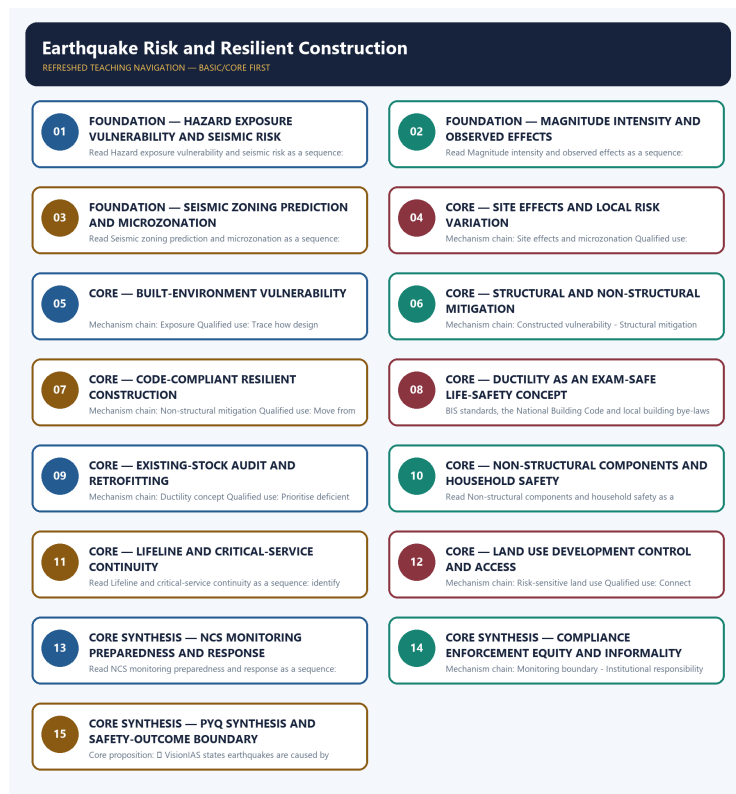


SOLVED PRACTICE WORKBOOK

Earthquake Risk and Resilient Construction - Learner-v2 Refreshed

UPSC complete learning session | Foundation to advanced | Concepts, evidence, practice and answer writing



Original deterministic study visual; labels are explanatory, not textual quotations.

Source order: repository knowledge -> official current linkage -> clearly marked analysis. No fabricated PYQs or statistics.

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Earthquake Risk and Resilient Construction - Solved Practice Workbook

BASIC MCQS / REMEDIATION

Q1. Which statement correctly identifies Seismic risk?

A. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. B. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. C. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. D. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction.

Answer: A. Explanation: Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q2. Which option preserves the risk or institutional boundary of Seismic risk?

A. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. B. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. C. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. D. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations.

Answer: B. Explanation: Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q3. Which statement uses Seismic risk without changing its hazard, mandate or status?

A. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. B. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. C. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. D. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning.

Answer: C. Explanation: Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q4. Which option avoids the standard UPSC close-option trap about Seismic risk?

A. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. B. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. C. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. D. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster.

Answer: D. Explanation: Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q5. Which statement correctly identifies Prediction boundary?

A. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. B. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. C. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. D. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning.

Answer: A. Explanation: Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q6. Which option preserves the risk or institutional boundary of Prediction boundary?

A. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. B. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. C. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. D. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings.

Answer: B. Explanation: Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q7. Which statement uses Prediction boundary without changing its hazard, mandate or status?

A. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. B. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. C. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. D. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure.

Answer: C. Explanation: Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q8. Which option avoids the standard UPSC close-option trap about Prediction boundary?

A. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. B. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. C. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. D. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction.

Answer: D. Explanation: Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q9. Which statement correctly identifies Magnitude and intensity?

A. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. B. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. C. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. D. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings.

Answer: A. Explanation: Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q10. Which option preserves the risk or institutional boundary of Magnitude and intensity?

A. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. B. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. C. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. D. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure.

Answer: B. Explanation: Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q11. Which statement uses Magnitude and intensity without changing its hazard, mandate or status?

A. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. B. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. C. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. D. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure.

Answer: C. Explanation: Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. The other options change the hazard or risk category, institutional owner, legal

Q12. Which option avoids the standard UPSC close-option trap about Magnitude and intensity?

A. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. B. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. C. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. D. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations.

Answer: D. Explanation: Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q13. Which statement correctly identifies Zoning and prediction?

A. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. B. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. C. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. D. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning.

Answer: A. Explanation: Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q14. Which option preserves the risk or institutional boundary of Zoning and prediction?

A. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. B. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. C. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. D. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure.

Answer: B. Explanation: Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q15. Which statement uses Zoning and prediction without changing its hazard, mandate or status?

A. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. B. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. C. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. D. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure.

Answer: C. Explanation: Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or

Q16. Which option avoids the standard UPSC close-option trap about Zoning and prediction?

A. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. B. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. C. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. D. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building.

Answer: D. Explanation: Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q17. Which statement correctly identifies Site effects and microzonation?

A. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. B. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. C. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. D. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure.

Answer: A. Explanation: Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q18. Which option preserves the risk or institutional boundary of Site effects and microzonation?

A. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. B. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. C. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. D. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment.

Answer: B. Explanation: Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q19. Which statement uses Site effects and microzonation without changing its hazard, mandate or status?

A. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. B. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. C. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. D. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained.

Answer: C. Explanation: Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. The other

options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q20. Which option avoids the standard UPSC close-option trap about Site effects and microzonation?

A. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. B. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. C. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. D. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning.

Answer: D. Explanation: Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q21. Which statement correctly identifies Exposure?

A. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. B. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. C. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. D. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents.

Answer: A. Explanation: Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q22. Which option preserves the risk or institutional boundary of Exposure?

A. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. B. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. C. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. D. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents.

Answer: B. Explanation: Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q23. Which statement uses Exposure without changing its hazard, mandate or status?

A. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. B. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. C. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. D. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents.

Answer: C. Explanation: Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q24. Which option avoids the standard UPSC close-option trap about Exposure?

A. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. B. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. C. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. D. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings.

Answer: D. Explanation: Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q25. Which statement correctly identifies Constructed vulnerability?

A. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. B. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. C. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. D. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents.

Answer: A. Explanation: Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q26. Which option preserves the risk or institutional boundary of Constructed vulnerability?

A. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. B. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. C. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. D. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents.

Answer: B. Explanation: Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q27. Which statement uses Constructed vulnerability without changing its hazard, mandate or status?

A. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. B. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. C. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. D. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure.

Answer: C. Explanation: Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q28. Which option avoids the standard UPSC close-option trap about Constructed vulnerability?

A. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. B. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. C. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. D. Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure.

Answer: D. Explanation: Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q29. Which statement correctly identifies Structural mitigation?

A. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. B. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. C. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. D. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure.

Answer: A. Explanation: Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q30. Which option preserves the risk or institutional boundary of Structural mitigation?

A. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. B. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. C. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. D. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete.

Answer: B. Explanation: Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage,

Q31. Which statement uses Structural mitigation without changing its hazard, mandate or status?

A. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. B. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. C. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. D. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure.

Answer: C. Explanation: Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q32. Which option avoids the standard UPSC close-option trap about Structural mitigation?

A. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. B. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. C. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. D. Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment.

Answer: D. Explanation: Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q33. Which statement correctly identifies Non-structural mitigation?

A. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. B. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. C. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. D. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete.

Answer: A. Explanation: Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q34. Which option preserves the risk or institutional boundary of Non-structural mitigation?

A. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. B. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. C. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. D. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing.

Answer: B. Explanation: Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. The other options change the hazard or risk category, institutional owner, legal instrument, warning

Q35. Which statement uses Non-structural mitigation without changing its hazard, mandate or status?

A. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. B. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. C. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. D. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention.

Answer: C. Explanation: Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q36. Which option avoids the standard UPSC close-option trap about Non-structural mitigation?

A. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. B. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. C. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. D. Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents.

Answer: D. Explanation: Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q37. Which statement correctly identifies Code-compliant design?

A. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. B. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. C. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. D. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing.

Answer: A. Explanation: BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q38. Which option preserves the risk or institutional boundary of Code-compliant design?

A. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. B. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. C. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. D. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing.

Answer: B. Explanation: BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built,

inspected or maintained. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q39. Which statement uses Code-compliant design without changing its hazard, mandate or status?

A. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. B. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. C. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. D. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention.

Answer: C. Explanation: BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q40. Which option avoids the standard UPSC close-option trap about Code-compliant design?

A. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. B. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. C. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. D. BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained.

Answer: D. Explanation: BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q41. Which statement correctly identifies Ductility concept?

A. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. B. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. C. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. D. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention.

Answer: A. Explanation: Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q42. Which option preserves the risk or institutional boundary of Ductility concept?

A. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. B. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. C. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. D. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing.

Answer: B. Explanation: Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q43. Which statement uses Ductility concept without changing its hazard, mandate or status?

A. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. B. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. C. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. D. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard.

Answer: C. Explanation: Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q44. Which option avoids the standard UPSC close-option trap about Ductility concept?

A. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. B. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. C. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. D. Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure.

Answer: D. Explanation: Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q45. Which statement correctly identifies Retrofitting boundary?

A. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. B. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. C. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. D. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention.

Answer: A. Explanation: Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage,

evidence rung or implementation status.

Q46. Which option preserves the risk or institutional boundary of Retrofitting boundary?

A. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. B. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. C. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. D. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention.

Answer: B. Explanation: Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q47. Which statement uses Retrofitting boundary without changing its hazard, mandate or status?

A. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. B. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. C. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. D. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard.

Answer: C. Explanation: Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q48. Which option avoids the standard UPSC close-option trap about Retrofitting boundary?

A. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. B. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. C. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. D. Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete.

Answer: D. Explanation: Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q49. Which statement correctly identifies Non-structural components?

A. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. B. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. C. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. D. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone.

Answer: A. Explanation: Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. The other options

change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q50. Which option preserves the risk or institutional boundary of Non-structural components?

A. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. B. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. C. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. D. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard.

Answer: B. Explanation: Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q51. Which statement uses Non-structural components without changing its hazard, mandate or status?

A. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. B. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. C. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. D. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event.

Answer: C. Explanation: Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q52. Which option avoids the standard UPSC close-option trap about Non-structural components?

A. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. B. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. C. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. D. Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing.

Answer: D. Explanation: Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q53. Which statement correctly identifies Lifeline resilience?

A. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. B. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. C. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. D. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard.

Answer: A. Explanation: Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q54. Which option preserves the risk or institutional boundary of Lifeline resilience?

A. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. B. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. C. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. D. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event.

Answer: B. Explanation: Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q55. Which statement uses Lifeline resilience without changing its hazard, mandate or status?

A. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. B. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. C. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. D. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery.

Answer: C. Explanation: Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q56. Which option avoids the standard UPSC close-option trap about Lifeline resilience?

A. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. B. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. C. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. D. Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention.

Answer: D. Explanation: Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q57. Which statement correctly identifies Risk-sensitive land use?

A. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. B. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. C. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. D. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard.

Answer: A. Explanation: Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q58. Which option preserves the risk or institutional boundary of Risk-sensitive land use?

A. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. B. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. C. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. D. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event.

Answer: B. Explanation: Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q59. Which statement uses Risk-sensitive land use without changing its hazard, mandate or status?

A. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. B. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. C. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. D. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence.

Answer: C. Explanation: Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q60. Which option avoids the standard UPSC close-option trap about Risk-sensitive land use?

A. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. B. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. C. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. D. Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone.

Answer: D. Explanation: Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. The other options change the hazard or risk category, institutional owner, legal instrument,

warning stage, evidence rung or implementation status.

Q61. Which statement correctly identifies Compliance chain?

A. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. B. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. C. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. D. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action.

Answer: A. Explanation: Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q62. Which option preserves the risk or institutional boundary of Compliance chain?

A. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. B. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. C. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. D. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence.

Answer: B. Explanation: Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q63. Which statement uses Compliance chain without changing its hazard, mandate or status?

A. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. B. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. C. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. D. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence.

Answer: C. Explanation: Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q64. Which option avoids the standard UPSC close-option trap about Compliance chain?

A. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. B. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. C. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. D. Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard.

Answer: D. Explanation: Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. The other options change the hazard or risk category, institutional owner, legal instrument, warning

stage, evidence rung or implementation status.

Q65. Which statement correctly identifies Monitoring boundary?

A. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. B. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. C. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. D. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence.

Answer: A. Explanation: The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q66. Which option preserves the risk or institutional boundary of Monitoring boundary?

A. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. B. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. C. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. D. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster.

Answer: B. Explanation: The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q67. Which statement uses Monitoring boundary without changing its hazard, mandate or status?

A. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. B. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. C. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. D. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster.

Answer: C. Explanation: The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q68. Which option avoids the standard UPSC close-option trap about Monitoring boundary?

A. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. B. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. C. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. D. The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event.

Answer: D. Explanation: The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. The

other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q69. Which statement correctly identifies Institutional responsibility?

A. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. B. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. C. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. D. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery.

Answer: A. Explanation: Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q70. Which option preserves the risk or institutional boundary of Institutional responsibility?

A. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. B. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. C. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. D. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence.

Answer: B. Explanation: Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q71. Which statement uses Institutional responsibility without changing its hazard, mandate or status?

A. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. B. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. C. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. D. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations.

Answer: C. Explanation: Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q72. Which option avoids the standard UPSC close-option trap about Institutional responsibility?

A. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. B. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. C. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. D. Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action.

Answer: D. Explanation: Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q73. Which statement correctly identifies Equity and informality?

A. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. B. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. C. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. D. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence.

Answer: A. Explanation: Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q74. Which option preserves the risk or institutional boundary of Equity and informality?

A. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. B. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. C. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. D. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations.

Answer: B. Explanation: Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q75. Which statement uses Equity and informality without changing its hazard, mandate or status?

A. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. B. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. C. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. D. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations.

Answer: C. Explanation: Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q76. Which option avoids the standard UPSC close-option trap about Equity and informality?

A. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. B. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. C. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. D. Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery.

Answer: D. Explanation: Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q77. Which statement correctly identifies Safety-outcome firewall?

A. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. B. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. C. Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. D. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations.

Answer: A. Explanation: A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q78. Which option preserves the risk or institutional boundary of Safety-outcome firewall?

A. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. B. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. C. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. D. Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction.

Answer: B. Explanation: A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q79. Which statement uses Safety-outcome firewall without changing its hazard, mandate or status?

A. Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. B. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. C. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. D. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning.

Answer: C. Explanation: A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate

evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

Q80. Which option avoids the standard UPSC close-option trap about Safety-outcome firewall?

A. Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. B. Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. C. Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. D. A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence.

Answer: D. Explanation: A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. The other options change the hazard or risk category, institutional owner, legal instrument, warning stage, evidence rung or implementation status.

PYQS AND ANSWER PRACTICE

AUDITED TOPIC-SPECIFIC PYQ OWNERSHIP

The 2021 GS-III card is directly routed. The 2019 vulnerability and 2024 resilience cards are explicitly conservative cross-topic applications and do not displace Topic 01 ownership.

PYQ DEMAND CARD 1 - 2021 GS-III

Demand: Discuss India's vulnerability to earthquake hazards and use historical disaster examples.

Status: Verified direct routing: Discuss · 10 marks · 150 words; examples must remain source-bounded and need no casualty or magnitude recital.

Model solution: Seismic risk: Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. **Zoning and prediction:** Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. **Site effects and microzonation:** Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. **Exposure:** Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. **Constructed vulnerability:** Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. **Non-structural mitigation:** Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. **Risk-sensitive land use:** Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. **Equity and informality:** Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 2 - 2019 GS-III

Demand: Discuss vulnerability as a concept for defining disaster impacts and explain its types.

Status: Verified direct ownership remains Topic 01; this conservative card supplies the earthquake site, construction, lifeline and social-vulnerability application.

Model solution: Seismic risk: Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. **Site effects and microzonation:** Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. **Exposure:** Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. **Constructed vulnerability:** Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. **Non-structural components:** Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. **Lifeline resilience:** Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. **Equity and informality:** Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

PYQ DEMAND CARD 3 - 2024 GS-III

Demand: Describe the elements that determine disaster resilience.

Status: Verified direct ownership remains Topic 01; this card routes code compliance, ductility, retrofit, land use and lifeline continuity as seismic-resilience elements.

Model solution: Structural mitigation: Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment.

Non-structural mitigation: Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents.

Code-compliant design: BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. **Ductility concept:** Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure.

Retrofitting boundary: Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. **Lifeline resilience:** Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. **Compliance chain:** Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. **Safety-outcome firewall:** A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. The solution therefore answers the routed demand without inventing an official model answer or an objective answer key.

ORIGINAL MAINS 1 - 10 MARKS

Question: Distinguish earthquake hazard, exposure, vulnerability and seismic risk. Answer in about 150 words.

Model thesis: Claim: Seismic risk. **Named evidence/example:** Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Prediction boundary. **Named evidence/example:** Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction.

Analysis: This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before

drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure. **Named evidence/example:** Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Constructed vulnerability. **Named evidence/example:** Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster.
- Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction.
- Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings.
- Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure.

Qualified conclusion: **Claim:** Seismic risk. **Named evidence/example:** Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Prediction boundary.

Named evidence/example: Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction.

Analysis: This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Exposure. **Named evidence/example:** Dense settlements, informal housing, schools, hospitals, bridges, utilities and other lifelines create concentrated exposure when located in seismic-risk settings. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Constructed vulnerability. **Named evidence/example:** Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 2 - 10 MARKS

Question: Differentiate magnitude, intensity, seismic zoning and prediction. Answer in about 150 words.

Model thesis: **Claim:** Prediction boundary. **Named evidence/example:** Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Magnitude and intensity. **Named evidence/example:** Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date,

institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Zoning and prediction. **Named evidence/example:** Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Site effects and microzonation. **Named evidence/example:** Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction.
- Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations.
- Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building.
- Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning.

Qualified conclusion: **Claim:** Prediction boundary. **Named evidence/example:** Earthquake occurrence cannot be precisely predicted by magnitude, place and time, so risk reduction must concentrate on vulnerability, preparedness and continuity rather than deterministic prediction. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Magnitude and intensity. **Named evidence/example:** Magnitude describes the size or energy of an earthquake event, whereas intensity describes observed effects at a place; one event has one reported magnitude but can produce different intensities across locations. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Zoning and prediction. **Named evidence/example:** Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Site effects and microzonation. **Named evidence/example:** Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 3 - 15 MARKS

Question: Explain structural and non-structural measures for earthquake-risk reduction. Answer in about 250 words.

Model thesis: **Claim:** Structural mitigation. **Named evidence/example:** Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Non-structural mitigation. **Named evidence/example:** Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung

before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Code-compliant design. **Named evidence/example:** BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Ductility concept. **Named evidence/example:** Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Non-structural components. **Named evidence/example:** Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Risk-sensitive land use. **Named evidence/example:** Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment.
- Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents.
- BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained.
- Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure.
- Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing.
- Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone.

Qualified conclusion: **Claim:** Structural mitigation. **Named evidence/example:** Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Non-structural mitigation. **Named evidence/example:** Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Code-compliant design. **Named evidence/example:** BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Ductility concept. **Named evidence/example:** Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. **Analysis:** This fixes the hazard, risk or

protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Non-structural components. **Named evidence/example:** Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Risk-sensitive land use. **Named evidence/example:** Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 4 - 15 MARKS

Question: Analyse the priorities and limits of seismic retrofitting and lifeline resilience. Answer in about 250 words.

Model thesis: **Claim:** Retrofitting boundary. **Named evidence/example:** Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Lifeline resilience. **Named evidence/example:** Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Institutional responsibility. **Named evidence/example:** Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Equity and informality. **Named evidence/example:** Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Safety-outcome firewall. **Named evidence/example:** A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete.
- Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention.
- Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action.
- Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery.
- A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence.

Qualified conclusion: Claim: Retrofitting boundary. **Named evidence/example:** Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Lifeline resilience.

Named evidence/example: Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Institutional responsibility. **Named evidence/example:** Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Equity and informality. **Named evidence/example:** Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Safety-outcome firewall. **Named evidence/example:** A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 5 - 20 MARKS

Question: Critically examine why code-compliant construction is primarily a governance and enforcement challenge. Answer in about 300 words.

Model thesis: Claim: Constructed vulnerability. **Named evidence/example:** Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Structural mitigation. **Named evidence/example:** Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Non-structural mitigation. **Named evidence/example:** Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Code-compliant design. **Named evidence/example:** BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Ductility concept. **Named evidence/example:** Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the

mandate-to-implementation-to-outcome boundary. **Claim:** Compliance chain. **Named evidence/example:** Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Institutional responsibility. **Named evidence/example:** Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Equity and informality. **Named evidence/example:** Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Safety-outcome firewall. **Named evidence/example:** A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure.
- Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment.
- Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents.
- BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained.
- Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure.
- Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard.
- Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action.
- Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery.
- A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence.

Qualified conclusion: **Claim:** Constructed vulnerability. **Named evidence/example:** Irregular form, weak materials, poor connections, deficient workmanship, unauthorised alteration and absent maintenance can turn ground shaking into collapse and service failure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Structural mitigation. **Named evidence/example:** Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Non-structural mitigation. **Named evidence/example:** Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. **Analysis:** This fixes the hazard, risk

or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Code-compliant design. **Named evidence/example:** BIS standards, the National Building Code and local building bye-laws form a safety framework, but a published code is not proof that a particular structure was correctly designed, built, inspected or maintained. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Ductility concept. **Named evidence/example:** Ductility is the capacity to undergo deformation while retaining life-safety resistance; for UPSC purposes it explains why controlled energy dissipation matters, not how to calculate or detail a structure. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Compliance chain. **Named evidence/example:** Seismic safety depends on linked code adoption, competent design, trained construction, approval, inspection, maintenance and enforcement; failure at one link can defeat the formal standard. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Institutional responsibility. **Named evidence/example:** Central and State authorities set standards and protect structures under their control, while States and local bodies identify, regulate and prioritise vulnerable or lifeline stock for action. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Equity and informality. **Named evidence/example:** Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Safety-outcome firewall. **Named evidence/example:** A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

ORIGINAL MAINS 6 - 20 MARKS

Question: Design an exam-safe earthquake-resilience framework covering land use, construction, retrofitting, lifelines, preparedness and equity. Answer in about 300 words.

Model thesis: **Claim:** Seismic risk. **Named evidence/example:** Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Zoning and prediction. **Named evidence/example:** Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Site effects and microzonation. **Named evidence/example:** Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Structural mitigation. **Named evidence/example:** Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy

implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Non-structural mitigation. **Named evidence/example:** Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Retrofitting boundary. **Named evidence/example:** Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Non-structural components. **Named evidence/example:** Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Lifeline resilience. **Named evidence/example:** Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Risk-sensitive land use. **Named evidence/example:** Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring boundary. **Named evidence/example:** The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Equity and informality. **Named evidence/example:** Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Safety-outcome firewall. **Named evidence/example:** A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.

Claim -> named evidence -> analysis -> qualification:

- Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster.
- Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building.
- Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning.
- Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment.
- Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents.

- Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete.
- Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing.
- Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention.
- Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone.
- The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event.
- Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery.
- A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence.

Qualified conclusion: Claim: Seismic risk. **Named evidence/example:** Earthquake risk arises when seismic hazard interacts with exposed people and assets, vulnerable sites and construction, and limited coping capacity; the event alone is not the disaster. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Zoning and prediction.

Named evidence/example: Seismic zoning classifies broad expected hazard for planning and design; it neither predicts the next earthquake nor determines the fate of an individual building. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Site effects and microzonation. **Named**

evidence/example: Local ground conditions, slope, soil and built form can modify shaking and secondary effects, so microzonation refines broad regional zoning for risk-sensitive local planning. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Structural mitigation. **Named evidence/example:**

Structural mitigation includes earthquake-resistant new construction and selective strengthening or retrofitting of deficient priority structures under competent professional assessment. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Non-structural mitigation. **Named evidence/example:**

Non-structural mitigation includes land-use control, code enforcement, professional training and licensing, safety audits, awareness, drills, emergency planning and securing hazardous contents. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Retrofitting boundary. **Named evidence/example:**

Retrofitting seeks to improve the safety of existing deficient structures after assessment and prioritisation; identification, sanction or audit is not proof that strengthening is complete. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication.

Qualification: The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Non-structural components. **Named evidence/example:**

Falling fixtures, equipment, partitions, storage and utility connections can injure occupants or disable services even when the main structural frame remains standing. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:**

Lifeline resilience. **Named evidence/example:** Hospitals, emergency facilities, transport links, water, power and communications require both physical safety and continuity arrangements because post-event functionality is a separate test from collapse prevention. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the

owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Risk-sensitive land use. **Named evidence/example:** Land-use decisions should avoid compounding shaking, slope, access and emergency-response risks; zoning must be connected to enforceable development control rather than treated as a map alone. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Monitoring boundary. **Named evidence/example:** The National Centre for Seismology monitors and reports earthquake parameters and supports hazard assessment; post-event parameter dissemination is not prediction of a future event. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Equity and informality. **Named evidence/example:** Lower-income, rural and peri-urban households may depend more on informal or owner-built construction and have fewer resources for audit, retrofit, insurance, relocation and recovery. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary. **Claim:** Safety-outcome firewall. **Named evidence/example:** A hazard map, code, guideline, audit, app, training programme or retrofit list proves an input; verified compliance, service continuity, reduced collapse and equitable recovery require separate evidence. **Analysis:** This fixes the hazard, risk or protection category, institutional mandate and verified evidence rung before drawing the policy implication. **Qualification:** The conclusion retains the owner's date, institution, legal character and the mandate-to-implementation-to-outcome boundary.